

Smart IOT Automated Water Hydration System

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Abstract—This paper presents the design and implementation of a Smart IoT-based Automated Water Hydration System for plant maintenance using the ESP32 microcontroller. The system integrates multiple environmental sensors—including an HC-SR04 ultrasonic sensor for water-level measurement, a DHT11 sensor for temperature and humidity sensing, a capacitive soil-moisture sensor, and a rain sensor—to ensure real-time monitoring of plant health. A servo-controlled shed mechanism provides protection during rainfall, while a relay-controlled hydration pump automatically waters the plant based on sensor-driven decision logic. The Blynk IoT platform enables live monitoring, user overrides, historical data visualization, and remote actuation. Experimental results show stable sensor performance, reliable cloud communication, and responsive automation suitable for home gardening and small-scale smart agriculture. The system demonstrates an efficient, low-cost, and scalable solution for autonomous plant hydration.

Index Terms—IoT, ESP32, Blynk, soil moisture, environmental monitoring, automated hydration, ultrasonic water level, smart agriculture.

I. INTRODUCTION

IoT-enabled automation systems are becoming increasingly important in agriculture and environmental monitoring. Smart plant-care systems reduce manual maintenance effort and enable optimized water usage through sensing technologies. This work documents the design, development, and evaluation of an IoT-based automated water hydration system that monitors environmental factors and controls plant hydration using a pump.

- 1) to enable real-time environmental monitoring using low-cost sensors,
- 2) to automate plant hydration based on intelligent decision logic, and
- 3) to provide remote monitoring and manual overrides using an IoT dashboard.

A. Background and Rationale

Smart agriculture and autonomous plant-care systems have gained significant importance due to the increasing need for efficient water management, labor reduction, and continuous real-time monitoring of environmental parameters. Manual plant hydration can be inconsistent and resource-intensive, especially in home gardening or small-scale agricultural environments. Existing hydration systems often lack automation, real-time monitoring, and adaptive decision-making capabilities based on multiple environmental factors. To address these limitations, this project proposes an IoT-enabled automated hydration system capable of sensing temperature, humidity, soil

moisture, rainfall, and water-tank status while autonomously regulating a hydration pump. The rationale behind this work is to develop a low-cost, scalable solution that ensures optimal plant hydration and improves resource efficiency through data-driven actuation.

B. Scope

The scope of this project is limited to the development, testing, and evaluation of a prototype-level IoT-based automated plant hydration system using low-cost, commercially available components such as the ESP32, ultrasonic water-level sensor, soil-moisture sensor, DHT11 module, rain sensor, servo motor, and relay-based pump controller. The system focuses on:

- 1) real-time environmental monitoring (temperature, humidity, rainfall, soil moisture),
- 2) automated hydration control through sensor-dependent pump activation,
- 3) safety mechanisms such as water-tank protection and pump-runtime limits,
- 4) remote monitoring and manual override using the Blynk IoT platform,
- 5) automatic shed operation based on rain detection.

This work does not cover advanced machine-learning-based moisture prediction, large-scale agricultural irrigation systems, multi-node wireless mesh deployment, solar-power optimization, or industrial-grade sensing technologies. The project is intended as a functional prototype that demonstrates the feasibility of low-cost automated hydration solutions for home gardening and small-scale smart agriculture applications.

II. RELATED WORK

Past IoT-based plant monitoring systems typically implement basic soil-moisture sensing or simple relay-based watering. Prior works involving ESP32 with Blynk demonstrate efficient IoT dashboards for remote monitoring. Studies on capacitive soil-moisture sensors show improved accuracy over resistive probes, while ultrasonic water-level sensing is widely adopted for tank measurement.

However, most existing low-cost systems lack combined environmental sensing, rainfall detection, intelligent hydration logic, and mechanical shed control. This project integrates these features into a unified automation model.

III. METHODOLOGY

A. Overview

The system consists of three main subsystems:

- **Sensing Unit:** Soil,Rain, Temperature–Humidity,Water Level
- **Actuation Unit:**Hydration Pump,Servo-controlled Shed
- **IoT Communication Unit:** Blynk Cloud Dashboard

Sensors continuously collect data from the environment. Based on preset thresholds, the hydration pump is activated only when necessary. The shed servo closes during rainfall to protect the plant. All data are transmitted to the Blynk mobile application for monitoring and manual control.

B. System Architecture

figure:1 demonstrates the IoT-based Automated Plant Hydration System. The ESP32 microcontroller serves as the central processing unit, collecting data from sensors monitoring soil moisture, water level, temperature, humidity, and rainfall. Based on these inputs, it autonomously controls a relay-operated pump for irrigation and a servo-driven shed for rain protection. Cloud integration via the Blynk platform enables real-time monitoring, remote control, and event logging, ensuring reliable autonomous operation with full oversight.

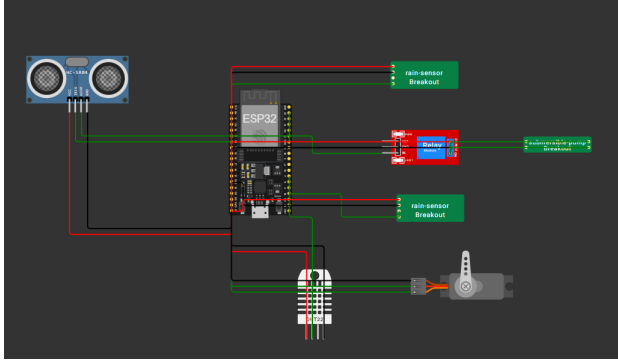


Fig. 1. System architecture of the IoT-based Automated Plant Hydration System.

TABLE I
PRIMARY HARDWARE COMPONENTS

Component	Role	Qty
ESP32 Dev Board	Main controller & IoT node	1
HC-SR04	Water-level sensing	1
DHT11	Temperature & humidity	1
Capacitive Soil Sensor	Soil moisture detection	1
Rain Sensor	Rainfall detection	1
Servo Motor	Shed motion	1
2-Channel Relay	Pump control	1
Water Pump	Hydration mechanism	1

C. Cloud Communication

The Blynk IoT platform was selected for its easy provisioning, intuitive mobile dashboard, and event-logging capabilities. The ESP32 transmits real-time sensor data—including soil moisture, water level, temperature, humidity, and rainfall—via Wi-Fi to the Blynk cloud, enabling remote monitoring. Users can also control the hydration pump, adjust thresholds, and override automation through the dashboard. Event logs and

notifications provide historical data and alerts for low water levels or sensor faults, ensuring reliable remote access, system oversight, and responsive plant hydration.

D. Hydration Logic and Environmental Processing

The system evaluates sensor inputs and governs plant hydration based on the following key points:

- **Sensor Monitoring:** Continuously reads soil moisture, temperature, humidity, water level, and rainfall.
- **Soil Dryness:** Determined when the capacitive soil sensor reading exceeds the predefined threshold.
- **Environmental Assessment:**
 - High heat flagged when temperature $\geq 30^{\circ}\text{C}$.
 - Low atmospheric moisture detected when humidity $\leq 40\%$.
- **Water Availability:** Ultrasonic sensor measures reservoir distance; distance > 12 cm indicates low water level, preventing pump activation.
- **Irrigation Decision:** Pump operates only if:
 - Soil is dry,
 - Environmental conditions require irrigation, and
 - Sufficient water is available.
- **Pump Safety:** Runtime limited to 10 seconds per cycle to prevent over-hydration and hardware strain.
- **Rain Response:** Servo-controlled shed automatically closes during rainfall to protect the plant.
- **Manual Overrides:** Users can control both pump and shed via the Blynk dashboard independently of automatic logic.

E. Control and Operation Sequence

The automated plant hydration system as shown in figure:2 follows a structured process to ensure autonomous operation. Sensors for soil moisture, temperature, humidity, water level, and rain are first initialized, along with the servo for the shed and the relay for the pump. The ESP32 connects to Wi-Fi and the Blynk platform for remote monitoring and control. The system continuously reads sensor data and transmits it to the Blynk dashboard. Environmental conditions—including soil dryness, high temperature, low humidity, and water availability—are evaluated to decide whether irrigation is required. The shed servo is adjusted automatically during rainfall to protect the plant, while manual overrides via Blynk remain available. Based on these decisions, the pump and shed are actuated accordingly. Sensor readings, system states, and events such as pump activation or low-water warnings are logged. This cycle of sensing, evaluation, actuation, and monitoring repeats at regular intervals, maintaining reliable autonomous hydration with full oversight.

IV. IMPLEMENTATION

A. Microcontroller Programming

The ESP32 firmware was developed using the Arduino environment, leveraging its extensive hardware abstraction

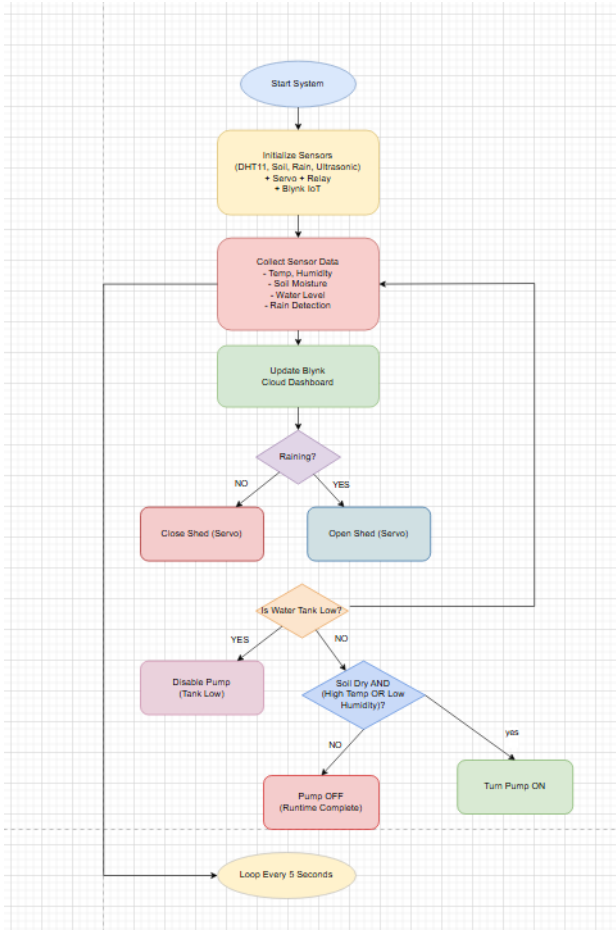


Fig. 2. Flowchart of the control and operation

This mapping supports modular data handling and simplifies integration with the Blynk dashboard.

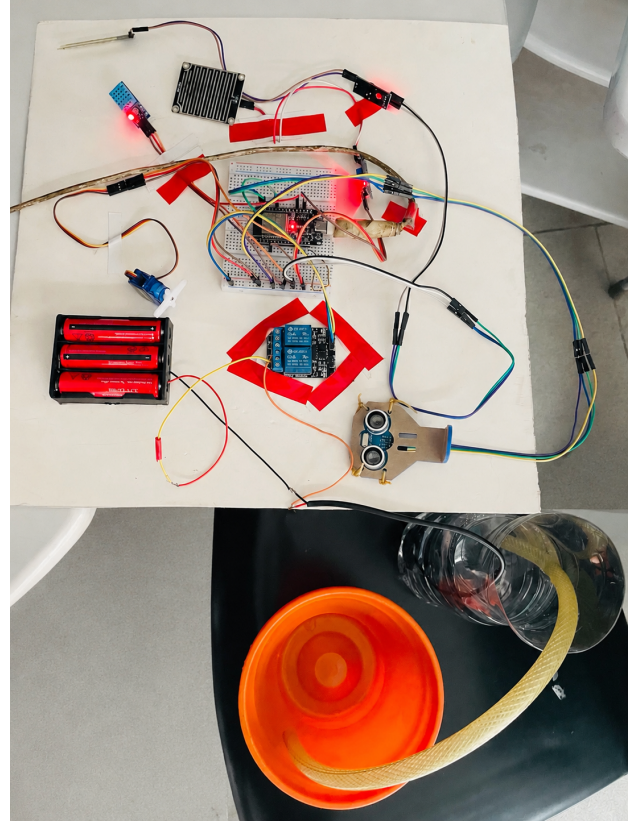


Fig. 3. system prototype

and library support. At startup, the program initializes all connected sensors, configures GPIO pins for relay and servo actuation, and establishes a Wi-Fi connection for Blynk IoT communication. A scheduled timer task executes the system's core processing loop every five seconds, ensuring periodic sensor polling and timely execution of hydration and shed-control logic.

B. Sensor Integration

Each sensing module in the system was individually validated and calibrated to ensure accurate operation under varying environmental and lighting conditions. The ESP32 collects raw sensor data and forwards the processed values to the Blynk IoT platform using dedicated virtual pins for structured data management. The virtual-pin assignments are summarized as follows:

- Soil moisture level → **V1**
- Water-tank distance (ultrasonic) → **V2**
- Rain detection (digital) → **V3**
- Pump status and manual control → **V4**
- Shed servo position → **V5**
- Temperature and humidity (DHT11) → **V0** and **V6**

C. Hydration Pump Operation

The hydration pump is controlled via a low-trigger relay to provide safe and electrically isolated switching. The system integrates multiple safeguards to prevent unintended activation:

- The pump operates only when the environmental thresholds are satisfied.
- Manual override commands from the IoT dashboard take priority over the automatic logic.
- All pump start and stop events are logged on Blynk for operational tracking and diagnostics.

These mechanisms collectively enhance the reliability of the system and ensure user accountability.

D. Software Components and Libraries

The IoT-based automated hydration system relies on several software components and libraries that enable sensor acquisition, actuator control, and cloud connectivity. The ESP32 firmware is developed using the Arduino framework for its extensive hardware support and lightweight runtime. Key libraries include:

- **Blynk IoT Library:** Provides cloud connectivity, virtual pin communication, event logging, and dashboard integration for real-time monitoring and manual overrides.

- **WiFi.h:** Enables Wi-Fi initialization and secure communication with the Blynk cloud.
- **DHT Sensor Library:** Facilitates reliable data acquisition from the DHT11 sensor with built-in error handling.
- **ESP32Servo Library:** Supports precise PWM control of the servo motor for weather-adaptive shed movement.
- **BlynkTimer:** Handles periodic sensor-reading tasks to ensure non-blocking execution.

These components ensure accurate sensor sampling, seamless wireless communication, and dependable real-time actuation across varying environmental conditions.

E. Evaluation Protocol

A structured testing protocol was executed to validate the functional stability and robustness of the system. Key evaluation steps included:

- Verification of soil-moisture readings under varying soil conditions.
- Assessment of ultrasonic water-level measurements at different tank depths.
- Analysis of rain-sensor response time and accuracy.
- Functional testing of the servo-based shed mechanism during simulated rainfall.
- Evaluation of pump activation under diverse temperature, humidity, and soil-moisture scenarios.
- Validation of IoT dashboard synchronization, remote commands, and system-status updates.

V. RESULTS AND DISCUSSION

A. Operational Results

The system successfully demonstrated the following functionalities:

- Stable real-time sensor monitoring.
- Automatic hydration based on multi-factor logic.
- Reliable ultrasonic water-level detection.
- Weather-responsive shed operation.
- Smooth servo motion and quick rainfall reaction.
- Effective Blynk integration for visualization and control.

B. Limitations and Challenges

The system exhibited certain limitations and challenges during testing:

- The DHT11 sensor shows delayed humidity response.
- The HC-SR04 ultrasonic sensor is sensitive to tank surface turbulence.
- Soil-moisture readings vary depending on soil composition.
- Wi-Fi connectivity affects real-time updates on the IoT dashboard.

C. System Learnings

Through the design, implementation, and testing of the system, several key lessons were identified. Proper sensor calibration is crucial to ensure accurate and consistent readings, particularly for soil moisture and environmental sensors. Maintaining a stable voltage supply significantly improves the

reliability of both sensors and actuators, preventing unexpected behavior or data fluctuations. Network reliability plays a vital role in ensuring timely updates and smooth operation of the IoT dashboard. Furthermore, implementing software timers helped reduce blocking delays in the main program loop, resulting in improved system responsiveness and more efficient execution of concurrent tasks. These insights will inform future improvements and optimizations in similar IoT-based automated systems.

VI. CONCLUSION AND FUTURE WORK

This paper presents a comprehensive IoT-based automated plant hydration system that integrates sensor-driven decision-making, cloud-based communication, and mechanical actuation. The developed prototype successfully achieves the primary objectives of efficient plant hydration, real-time environmental monitoring, and remote control through an intuitive IoT dashboard. The system demonstrated reliable performance under varying environmental conditions and provided timely responses to changes in soil moisture, water levels, and rainfall events.

For future work, several enhancements are proposed to further improve system intelligence and scalability. These include the integration of machine-learning algorithms for predictive hydration scheduling, expansion of the system to support multi-plant setups with centralized monitoring, and the incorporation of detailed water-usage analytics through cloud dashboards. Additional improvements may focus on sensor fusion, energy efficiency, and enhanced network reliability to ensure robust operation in diverse environments. These extensions aim to make the system more adaptive, efficient, and user-friendly for large-scale or commercial applications.

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